

SUSTAINABILITY NOTE

Circular PET Recovery with Transparent Impact Boundaries

A concise account of Poly Cleaner LLC's circular material activity, process-water reuse, Plastic Credit profile and approach to greenhouse-gas measurement.

CIRCULAR MATERIAL

Post-consumer PET converted into hot-washed rPET flakes.

WATER REUSE

Compact on-site treatment supports process-water recirculation.

TRACEABLE IMPACT

Plastic collection and recycling activity documented under Verra.

~90m

PET bottle-equivalents per year

Company activity figure; mass conversion requires a declared bottle-weight assumption.

20 m³/day

process-water reuse capacity

Up to 6,000 m³ at 300 operating days; actual reuse should be metered.

5,170

Plastic Credits — first issuance

Plastic Credits are plastic-impact units, not carbon credits or tCO₂e.

01 MATERIAL CIRCULARITY

Bottle-equivalents are a clear activity metric; carbon impact requires a defined footprint boundary

Poly Cleaner recovers post-consumer PET and converts it into hot-washed rPET flakes. The climate section below is a transparent screening scenario, not a verified product or corporate carbon footprint.



ACTIVITY INDICATOR

~90 million

PET bottle-equivalents recycled into rPET flakes per year

Bottle count is intuitive, but it is not a mass measurement. A defensible climate claim should use weighbridge and production records, plus site-specific energy, transport, water-treatment, chemical and yield data.

Recommended KPI
 kg CO₂e per tonne of saleable rPET flakes, supported by an ISO 14067-aligned product footprint study.

ILLUSTRATIVE LCA REFERENCE SCREEN

A North American industry comparison reports approximately 59% lower greenhouse-gas emissions for rPET resin production than virgin PET resin. Published factors used here are 2.233 kg CO₂e/kg for virgin PET and 0.91 kg CO₂e/kg for recycled PET — a difference of 1.323 kg CO₂e/kg.

Assumed PET mass per bottle	90m bottle mass-equivalent	Reference GHG difference
20 g	1,800 t	~2,380 tCO ₂ e
25 g	2,250 t	~2,980 tCO ₂ e
30 g	2,700 t	~3,570 tCO ₂ e

Interpretation: the middle row is a disclosed 25 g/bottle scenario, not a measured Poly Cleaner result. The reference studies concern North American resin systems and recycled PET pellets; Poly Cleaner produces flakes in Azerbaijan. A final claim must replace all proxy data with a plant-specific inventory and consistent system boundary.

CLAIMS BOUNDARY | This note does not claim carbon neutrality, net zero, a verified product carbon footprint, or carbon credits. The screening values communicate order of magnitude only.

02 PROCESS-WATER REUSE

A compact treatment unit returns process water to use within the plant

Poly Cleaner reports treatment and reuse capacity of up to 20 m³ per day. This reduces demand for new water and can reduce the volume of process wastewater leaving the site, subject to operating conditions and water-quality controls.

ILLUSTRATIVE ANNUAL REUSE POTENTIAL

6,000 m³

equivalent to 6 million litres

20 m³/day × 300 operating days = 6,000 m³/year

Actual annual reuse depends on operating days, treatment uptime, water quality and measured flow volumes.

EXPECTED OPERATIONAL EFFECT

- 01** Lower demand for incoming freshwater
- 02** Lower process-effluent volume requiring discharge or off-site management
- 03** Greater resilience of the washing process during water-supply constraints
- 04** A measurable water-efficiency KPI for customers and future reporting

Measurement controls for a credible public water claim

Metric	Minimum evidence	Recommended disclosure
Water treated	Daily flow meter / unit log	m ³ treated per month and year
Water reused	Return-line meter or reconciled balance	% and m ³ of treated water reused
Freshwater intake	Incoming-water meter / invoices	m ³ intake per tonne of rPET flakes
Effluent leaving site	Discharge meter / removal records	m ³ discharged and destination
Water quality	Operating parameters and test records	Reuse specification and compliance status

REPORTING ALIGNMENT

GRI 303 supports disclosure of water and effluent impacts. ISO 14046 provides principles for a life-cycle-based water footprint; a simple reused-water volume is a useful efficiency KPI but is not, by itself, a complete water-footprint assessment.

03 VERIFIED PLASTIC IMPACT

Plastic Credit accounting and carbon accounting serve different purposes

Poly Cleaner's Plastic Credit profile supports traceable plastic collection and recycling outcomes. It must remain separate from greenhouse-gas claims and carbon-credit accounting.

POLY CLEANER PLASTIC CREDIT PROFILE

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Plastic Credits in the first issuance

The project profile covers documented plastic collection and recycling activity, supported by independent assessment and registry-based issuance records.

Important: one Plastic Credit represents one tonne of collected or recycled plastic waste under the applicable Verra credit type. It does not represent one tonne of CO₂e.

VERIFIED PLASTIC OUTCOMES

- Waste Collection Credits (WCCs): additional plastic collected and appropriately managed above baseline.
- Waste Recycling Credits (WRCs): additional plastic recycled above baseline.
- Independent audits, approved methodologies and registry tracking support program integrity.

Plastic Credits should be presented as traceable plastic-impact units — never relabelled as carbon savings.

Roadmap to a decision-grade Sustainability Note

NOW

Disclose activity metrics

Bottle-equivalents, credited plastic outcomes and water-reuse capacity, each with a clear evidence basis.

NEXT

Build the corporate GHG inventory

Use the GHG Protocol Corporate Standard for Scope 1, purchased electricity under Scope 2 and relevant Scope 3 categories.

THEN

Complete the product footprint

Use ISO 14067 with ISO 14040/14044 principles to quantify kg CO₂e per tonne of saleable rPET flakes.

VERIFY

Assure material claims

Independently review source data, calculation files, boundaries, emission factors and public wording before publishing a verified result.

Method and source notes

- Verra Plastic Waste Reduction Standard
- GHG Protocol Standards & Guidance
- ISO 14067:2018 — Carbon footprint of products
- ISO 14046:2014 — Water footprint
- GRI 303: Water and Effluents 2018
- NAPCOR rPET / virgin PET LCA comparison

Evidence basis | Company figures in this note are management-provided operating indicators. LCA factors are external reference values. Public figures should be refreshed annually and reconciled to production, water, energy and Plastic Credit records.